

Terrain & Contour Modeling Demo

Elevation Contour Mapping
Islamabad / Kotli Region

This report documents a short demonstration of terrain elevation modeling using contour-based visualization. Elevation bands are color-coded at a 5-meter interval to highlight relief, low-lying basins, and surrounding topography around Islamabad and the Kotli / Azad Kashmir foothills.

ELEVATION CONTOURS

5-meter interval · Color-graduated relief bands
Generated Demo Report

Overview

This demo visualizes elevation contour data over two areas of interest: a rural stretch near Tikhsala / Kotli (Azad Kashmir) and the wider Islamabad–Rawalpindi metropolitan region. Elevation is represented using a graduated color scale, allowing quick visual identification of low-lying floodplains, mid-elevation plains, and higher terrain or hill ranges.

Color Legend (Elevation Bands, 5m Interval)

Color	Elevation Range
Dark Red	≥ 793 m
Orange	702 – 792 m
Yellow	611 – 701 m
Green	520 – 610 m
Blue	≤ 519 m

Method Summary

1. A digital elevation surface is sampled over the region of interest.
2. Elevation values are binned into 5-meter intervals.
3. Each bin is rendered as a color-filled contour band overlaid on the base map.
4. Users can click a contour band to inspect the underlying elevation value.

STEP 1

Area Select

Region of Interest Highlighted — Islamabad Capital Territory



The area of interest is selected on the base map by drawing a bounding box, here shown around the Islamabad Capital Territory footprint against the surrounding Margalla Hills and Rawalpindi urban area.

STEP 2

Contour Interval Set

Analysis Extent & Elevation Interval Configured (5m)

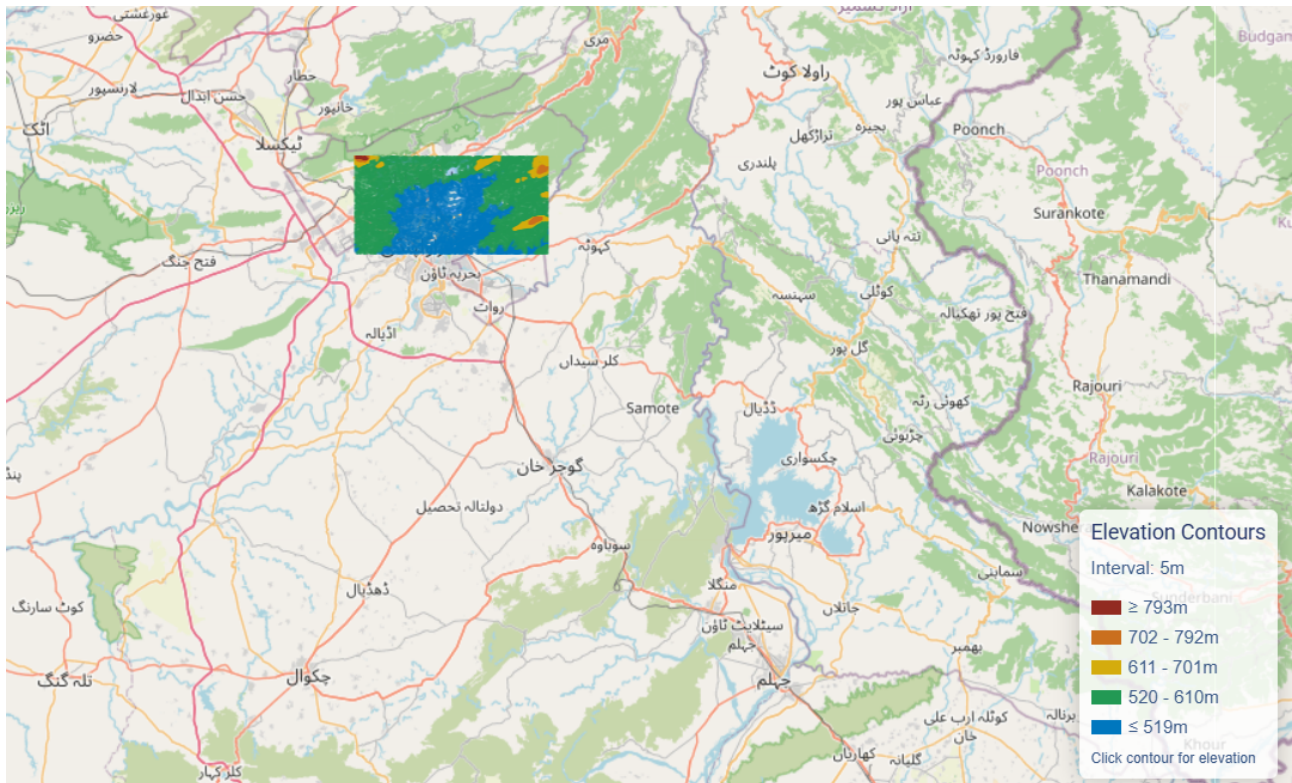


With the region locked in, the elevation sampling extent and contour interval (5 meters) are configured over the Islamabad / Rawalpindi base map ahead of generating the color-graduated elevation bands.

STEP 3

Final Contour Result

Elevation Contour Overlay — Kotli / Tikhsala Area



Final color-coded elevation contour output, overlaid on the base map near Tikhsala and Kotli, Azad Kashmir. The blue zone marks the lowest elevation band (≤ 519 m), transitioning through green, yellow, and orange toward the higher terrain at the patch edges, per the 5m interval legend.